

VIET-MAN LE

Project Assistant and PhD Candidate

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RESEARCH INTERESTS

Knowledge-Based Diagnosis, Conflict Detection, and Explanation; Machine Learning for Constraint Solving; Knowledge-Based Configuration and Software Product Lines; Constraint Acquisition; Large Language Models for Software Engineering.

EDUCATION

Graz University of Technology

Graz, Austria

PhD in Computer Science (in progress)

2020 – expected 2026

- Thesis: *Intelligent Techniques for Efficient Diagnostic Reasoning in Constraint-Based Systems*. Develops solver-agnostic algorithms for conflict detection, diagnosis, and quality assurance that scale to industrial constraint-based systems, such as configuration systems and recommender systems.
- Supervisor: Prof. Alexander Felfernig

Institute for Informatics in French-speaking areas

Hanoi, Vietnam

Master of Computer Science – Artificial Intelligence & Multimedia

2009 – 2011

- Double diplomas with University of La Rochelle (France) and Hanoi University of Science and Technology (Vietnam)
- Master's Thesis: *Système de navigation dans des bases de données d'images* (image browser based on Self-Organizing Maps). Supervisor: Prof. Michel Verleysen, Catholic University of Louvain, Belgium

University of Sciences, Hue University

Hue, Vietnam

Bachelor of Computer Science

2000 – 2004

SELECTED PUBLICATIONS

According to [Google Scholar](#) (May 2026): **775 citations**, h-index **13**, i10-index **17**, across more than 50 peer-reviewed papers. Full list at manleviet.github.io/publications.html.

Christian Bähnisch, Alexander Felfernig, Damian Garber, Albert Haag, Denis Helic, Lothar Hotz, **Viet-Man Le**, Sebastian Lubos. *Machine Learning for Constraint-based Configuration: A Survey*. *Journal of Artificial Intelligence Research* (to appear), 2026. **SCIMAGO Q1**

Alexander Felfernig, **Viet-Man Le**, Sebastian Lubos, Thi Ngoc Trang Tran. *Constraint Acquisition of Configuration Knowledge Bases with Maximum Satisfiable Subsets*. *Software and Systems Modeling* (under revision), 2026. **SCIMAGO Q1**

Viet-Man Le, Lukas A. Feldgrill, Alexander Felfernig. *Robust Lazy Conflict Detection via Multi-Conflict Extraction and Genetic Diversity Control*. *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, no. 23, pp. 19208–19215, 2026. **CORE A***

Viet-Man Le, Thi Ngoc Trang Tran, Sebastian Lubos, Alexander Felfernig, Damian Garber. *Early-Stage Product Line Validation Using LLMs: A Study on Semi-Formal Blueprint Analysis*. *Proceedings of the 41st ACM/SIGAPP Symposium on Applied Computing (SAC 2026)*, pp. 1620–1627, 2026. **CORE B**

Mathias Uta, **Viet-Man Le**, Alexander Felfernig, Denis Helic. *Learning Constraint Orderings for Direct Diagnosis*. *Journal of Intelligent Information Systems*, vol. 63, no. 5, pp. 1753–1777, 2025. **SCIMAGO Q2**

Viet-Man Le, Alexander Felfernig, Thi Ngoc Trang Tran, Mathias Uta. *InformedQX: Informed Conflict Detection for Over-Constrained Problems*. *Proceedings of the 38th AAAI Conference on Artificial Intelligence*, vol. 38, no. 9, pp. 10616–10623, 2024. **CORE A***

Viet-Man Le, Cristian Vidal Silva, Alexander Felfernig, David Benavides, José Galindo, Thi Ngoc Trang Tran. *FastDiagP: An Algorithm for Parallelized Direct Diagnosis*. *Proceedings of the 37th AAAI Conference on Artificial Intelligence*, vol. 37, no. 5, pp. 6442–6449, 2023. **CORE A***

Viet-Man Le, Alexander Felfernig, Mathias Uta, David Benavides, José Galindo, Thi Ngoc Trang Tran. *DirectDebug: Automated Testing and Debugging of Feature Models*. *Proceedings of the ACM/IEEE 43rd International Conference on Software Engineering: New Ideas and Emerging Results*, pp. 81–85, 2021. **CORE A***

EMPLOYMENT

Graz University of Technology	Graz, Austria
<i>University Assistant/Project Assistant and PhD Candidate</i>	2020 – present
- Designed and implemented solver-agnostic algorithms for conflict detection, direct diagnosis, redundancy detection, and automated debugging that scale to industrial constraint-based systems, such as configuration systems and recommender systems - Developed and maintained open-source research software (HiConfiT, DirectDebug, FM2ExConf) supporting feature model analysis and diagnosis, with contributions integrated into the flamapy ecosystem	
University of Economics, Hue University	Hue, Vietnam
<i>Vice-Dean, Faculty of Economic Information Systems</i>	2014 – 2019
- Led the design and launch of the Undergraduate Degree Program in Management Information Systems - Established faculty-level policies for academic advising, career planning, and student professionalisation - Coordinated curriculum development, teaching assignments, and quality assurance across the faculty's undergraduate programmes	
University of Economics, Hue University	Hue, Vietnam
<i>Lecturer, Faculty of Economic Information Systems</i>	2004 – 2020
- Taught Programming (C#, Python, Java), Data Structures and Algorithms, and Windows Applications Programming to undergraduate students - Researched recommender systems, with a focus on multi-agent models for collaborative filtering - Advised 2–6 undergraduate students per year (2012–2019) on bachelor theses and student research projects in information systems	

FUNDED RESEARCH PROJECTS

GenRE – Generative AI for Requirements Engineering	2024 – 2027
<i>FFG Bridge project. Team member. Industry partner: Morgendigital.</i>	
Develops LLM-based techniques for requirements elicitation, quality assurance, and validation in software engineering	
Kickscale: KI-Engine – Sales-Specialized LLM for the Kickscale SaaS Platform	2023 – 2025
<i>FFG Basisprogramm project. Scientific consulting role via TU Graz. Industry lead: Kickscale GmbH.</i>	
Developed LLM-based methods for extracting Ideal Customer Profiles from CRM deal data, and released frameworks for evaluating LLM and RAG outputs	
OpenSpace – AI Techniques for Testing Highly-Variant Software	2021 – 2024
<i>FFG Bridge project. Team member. Industry partner: Uniquare GmbH.</i>	
Developed machine-learning approaches for testing and debugging variability-intensive software, including automated analysis of variability models and identification of faulty components	
ParXCel – Scalable Constraint Solving via ML and Parallelization	2020 – 2023
<i>FFG Bridge project. Team member. Industry partner: Combeeneration GmbH.</i>	
Integrated machine learning into constraint-based reasoning and parallelized conflict detection and direct diagnosis to boost performance (FastDiagP, InformedQX)	

GRANTS & AWARDS

Projektfonds Lehre 2026 (<i>Project Fund for Teaching</i>)	2026
<i>TU Graz Vice-Rectorate for Teaching – MIRROR project (team member, PI: Prof. Alexander Felfernig)</i>	
Best Teaching Award – Nominations	2023, 2024, 2025
<i>Course Introduction to Structured Programming, Graz University of Technology</i>	
Young Experts: Minute Madness Session	2024
<i>Austrian Computer Science Day 2024 – session for excellent doctoral students in CS at Austrian universities</i>	
Hue University Research Grant	2015 – 2016
<i>Project: Multi-agent model for tourist attraction recommender systems (team member)</i>	

SERVICE

Conference and Workshop Organisation	
Chair, 27th International Workshop on Configuration (ConfWS 2025)	2025
<i>Co-located with the 28th European Conference on Artificial Intelligence (ECAI 2025), Bologna, Italy</i>	
Proceedings Chair, 26th ACM International Systems and Software Product Line Conference	2022
<i>Graz, Austria</i>	
Program Committee Member	
2025: RecSys 2025, SPLC 2025, ECAI 2025, AAAI 2026	
2024: ConfWS 2024, IntRS 2024, SPLC 2024	
Conference and Journal Reviewer	
2026: 1 × SoSyM, 1 × JIIS	
2025: 2 × RecSys 2025, 2 × SPLC 2025, 4 × ECAI 2025, 5 × AAAI 2026, 7 × JIIS	
2024: 2 × ConfWS 2024, 2 × IntRS 2024, 3 × SPLC 2024, 2 × ESWA	

TEACHING

Graz University of Technology (University Assistant)

Introduction to Structured Programming (ISP)	2020 – 2026
<i>Course organization, assignment design. Nominated for Best Teaching Award in 2023, 2024, 2025.</i>	
Object-Oriented Analysis and Design (OAD)	2022 – 2026
<i>Assignment design on design principles and patterns; lecturer on Design Patterns in 2024 and 2025.</i>	

University of Economics, Hue University (Lecturer, 2004 – 2019)

Courses taught: Introduction to Programming (C#, Python), Windows Applications Programming, Data Structures and Algorithms, Object-Oriented Programming.

MENTORING AND CO-SUPERVISION

Fabian True (Bachelor)	2025
<i>FM2ExConf: A Web-Service to Convert Feature Models into an Executable Representation in Microsoft Excel</i>	
Nhat Minh Hoang (Bachelor)	2025
<i>From Conflicts to Efficiency: Learning Constraint Orderings for QuickXPlain</i>	
Lukas A. Feldgrill (Master)	2025
<i>Robust Lazy Conflict Detection via Multi-Conflict Extraction and Genetic Diversity Control (AAAI 2026)</i>	
Yunus Emre Akca (Bachelor)	2023
<i>PWipeOutR: Parallelized Redundancy Detection for Feature Models</i>	
Tamim Burgstaller (Bachelor)	2022
<i>FMTesting: A FeatureIDE Plug-in for Automated Feature Model Analysis and Diagnosis (PAIS/ECAI 2023)</i>	

SOFTWARE & TOOLS

HiConfit – Suite of open-source Java libraries (*hiconfit-core*) and CLI apps (*KBStatistics*, *FMGen*) for Knowledge-Based Configuration Systems.

FMTesting – FeatureIDE plug-in for automated feature model testing and debugging (PAIS 2023).

FM2ExConf – Feature Model to Excel Configurator converter, built on hiconfit-core.

DirectDebug – Java software package for automated testing and debugging of variability models (Software Impacts, 2021).

flamapy – Contributed FastDiagP, DirectDebug, and WipeOutR plugins to the Python ecosystem for feature model analysis.

TECHNICAL SKILLS

Constraint Solvers & Tools: Choco, SAT, MiniZinc

Programming Languages: Java, Python, C/C++, SQL

Frameworks & DevOps: Spring Boot, FeatureIDE (Eclipse plug-in), Git, Docker, GitHub Actions

Languages: English (fluent), Vietnamese (native)

REFERENCES

Univ.-Prof. Dipl.-Ing. Dr.techn. Alexander Felfernig	PhD Supervisor
<i>Full Professor, Institute of Software Engineering and AI, Graz University of Technology</i>	
- Address: Inffeldgasse 16b/II, 8010 Graz, Austria - Email: alexander.felfernig@tugraz.at	
Dr. Habil. Binh Nguyen	Former Undergrad. Lecturer
<i>Research Scholar, International Institute for Applied Systems Analysis (IIASA), Austria</i>	
- Formerly Lecturer at the University of Sciences, Hue University, Vietnam - Address: Schlossplatz 1, 2361 Laxenburg, Austria - Email: nguyenb@iiasa.ac.at	